

UNIT-1

SET THEORY, RELATION AND FUNCTIONS

Definition (set)

A set is collection of well defined objects.

For example,

$$Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

is the set of all integers.

$$\text{or } I = \{x \mid x \text{ is an integer}\}$$

Definition: (Finite and Infinite Sets)

A set is said to be finite if it contains a finite number of distinguishable elements. Otherwise, a set is infinite.

Definition: Null set (or) Empty set (or) Void set

The set that has no elements in it is called a null set or an empty set.

It is denoted by the symbol ϕ or $\{\}$.

Definition: (Cardinality of a set)

Cardinality of a set is the number of elements in the set and it is denoted by $n(A)$ (or) $|A|$.

EX: If $A = \{1, 3, 5, 7\}$, then $n(A) = 4$.

Note: $|\phi| = 0$ but $\{0\} \neq \phi$. Also $\phi \neq \{\phi\}$.

Def: (Singleton set)

A set consisting of a single element is called a singleton set. i.e., $n(A) = 1$.

EX: $A = \{a\}$.

Def: (Subset)

A set 'A' is a subset of the set 'B' if and only if every element of 'A' is also an element of 'B'. We also say that 'A' is contained in 'B' and use the notation $A \subseteq B$.

Def: (Proper Subset)

A set 'A' is called proper subset of the set 'B' if, (i) A is subset of B, and
(ii) B is not a subset of A.

i.e., A is said to be a proper subset of B if every element of A belongs to the set B, but there is at least one element of B, which is not in A.

If A is a proper subset of B, then we denote it by $A \subset B$.

Def: (Superset)

If A is subset of B, then B is called a superset of A.

Def: (Universal set)

A set is called a universal set if it includes every set under discussion. A universal set is denoted by 'E' or 'U'.

Def: (Equality of sets)

Two sets A and B are said to be equal if and only if $A \subseteq B$ and $B \subseteq A$.

i.e., $A = B \iff (A \subseteq B) \text{ and } (B \subseteq A)$.

Def: (Power Set)

The set of all subsets of a set 'A' is called the Power set of A. It is denoted by $P(A)$ or 2^A .

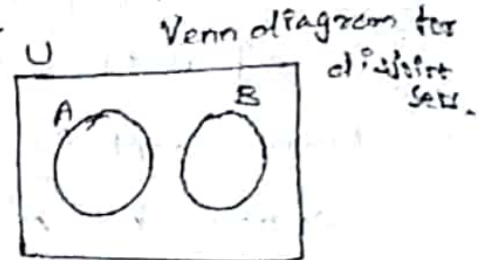
If A has 'n' elements in it, then $P(A)$ has 2^n elements.

Ex: Let $A = \{1, \phi\}$

Then $P(A) = \{\phi, \{1\}, \{1, \phi\}\}$.

Disjoint sets

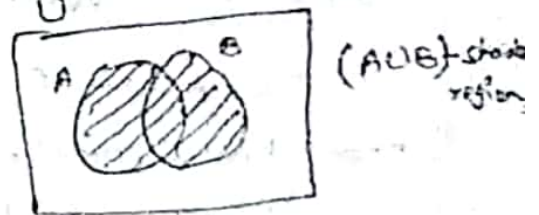
Two sets are said to be disjoint if they have no elements in common.



Union of two sets

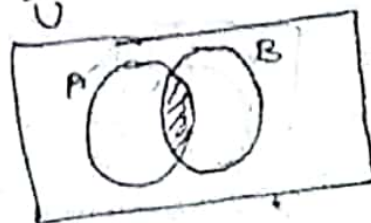
The Union of two sets A and B is the set whose elements are all of the elements in A or in B or in both. It is denoted by $(A \cup B)$.

$$\text{Thus } A \cup B = \{x \mid x \in A \text{ or } x \in B \text{ or } x \in A, B\}$$



Intersection of two sets

The Intersection of two sets A and B is the set whose elements are all of the elements common to both A and B. It is denoted by $(A \cap B)$.



$(A \cap B)$ - Shaded region

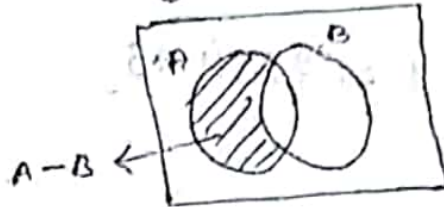
$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Difference of sets

If A and B are subsets of the universal set U , then the relative complement of B in A is the set of all elements in A which are not in B .

It is denoted by $A - B$.

$$\text{Thus } A - B = \{x / x \in A \text{ and } x \notin B\}$$



Complement of a set:

If U is a Universal set containing the set A , then $U - A$ is called the complement of A . It is denoted by A^c (or) A' (or) $\bar{A}$.

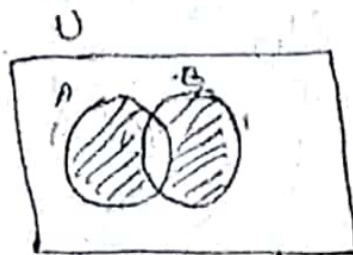
$$\text{Thus } A' = \{x / x \in A\}'$$



Symmetric difference of sets (Boolean sum of sets)

If A and B are subsets of the Universal set U , then the symmetric difference of A and B is the set of all elements that belong to either set but not to both. It is denoted by $A \Delta B$ (or) $A \oplus B$.

$$\text{Thus } A \Delta B = \{x / x \in A \text{ or } x \in B \text{ but } x \notin A \cap B\}$$



$$A \Delta B = (A - B) \cup (B - A)$$

Example

If A, B, C are any three sets, then prove that

$$(a) A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

$$(b) A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \text{ [Do yourself].}$$

Sol:

(a)

$$\begin{aligned} \text{Let } x \in A \cup (B \cap C) &\Rightarrow x \in A \text{ or } x \in B \cap C \\ &\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C) \\ &\Rightarrow (x \in A \text{ or } x \in B) \text{ and } \\ &\quad (x \in A \text{ or } x \in C) \\ &\Rightarrow x \in A \cup B \text{ and } x \in A \cup C \\ &\Rightarrow x \in (A \cup B) \cap (A \cup C) \end{aligned}$$

$$\therefore A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \longrightarrow (1)$$

Let

$$\begin{aligned} x \in (A \cup B) \cap (A \cup C) &\Rightarrow x \in A \cup B \text{ and } x \in A \cup C \\ &\Rightarrow (x \in A \text{ or } x \in B) \text{ and } \\ &\quad (x \in A \text{ or } x \in C) \\ &\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C) \\ &\Rightarrow x \in A \text{ or } x \in (B \cap C) \\ &\Rightarrow x \in A \cup (B \cap C) \end{aligned}$$

$$\therefore (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C) \longrightarrow (2)$$

From (1) and (2),

$$\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$$

Problem:

- ① How many natural numbers $n \leq 1000$ are not divisible by any of 2, 3?

Sol:

Let U be the universal set of all 1000 numbers.

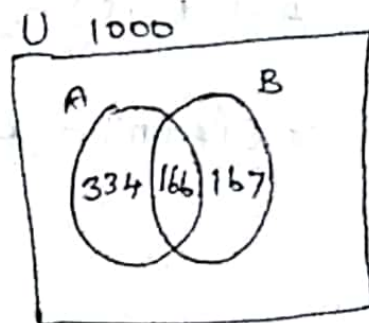
$$n(U) = 1000$$

A be the set of numbers divisible by 2
and B be the set of numbers divisible by 3.

$$n(A) = \frac{1000}{2} = 500$$

$$n(B) = \frac{1000}{3} = [333.\overline{3}] = 333$$

$$n(A \cap B) = \frac{1000}{2 \times 3} = 166$$



From the Venn diagram

The numbers that are not divisible by any of 2, 3 are $1000 - (334 + 167 + 166) = \underline{\underline{333}}$

- ② In a survey of 100 students, it was found that 40 studied mathematics, 64 studied physics, 35 studied chemistry, 1 studied all the three subjects, 25 studied mathematics and physics, 3 studied mathematics and chemistry and 20 studied physics and chemistry. Find the number of students who studied chemistry only and the number who studied none of these subjects.

(Do Yourself)

- ⑤ Among 100 students, 32 study mathematics, 20 study statistics, 45 study accountancy, 15 study mathematics and accountancy, 7 study mathematics and statistics, 10 study statistics and accountancy and 30 do not study any of the three subjects. Find (i) the number of students studying all three subjects (ii) the number of students studying exactly one of the three subjects.

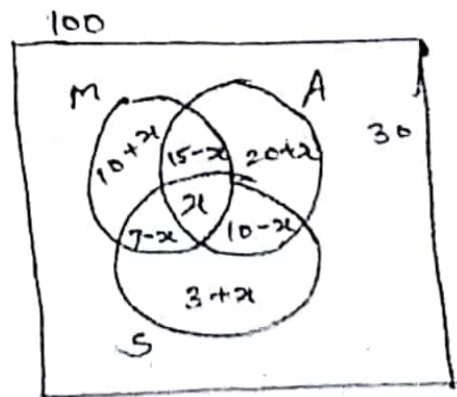
Sol:

let $n(E) = 100$

m denotes the set of students studying mathematics
 S " " " " " " statistics
 A " " " " " " accountancy.

Then $n(M) = 32$, $n(S) = 20$, $n(A) = 45$.

$n(M \cap A) = 15$, $n(M \cap S) = 7$ & $n(S \cap A) = 10$



Also 30 students do not study any of these subjects
 Hence

$$\begin{aligned}
 &10+x + 3+x + 20+x + 15-x \\
 &+ 7-x + 10-x + x + 30 = 100 \\
 &95 + x = 100 \\
 &x = 100 - 95
 \end{aligned}$$

$$x = 5$$

∴ 5 students study all three subjects

No. of sty. studying mathematics only

$$= 10 + 20 = 10 + 5 = 15$$

No. of sty. studying statistics only

$$= 3 + 20 = 3 + 5 = 8$$

No. of sty. studying Accountancy only

$$= 20 + 20 = 20 + 5 = 25$$

∴ No. of students studying only one subject

$$= 15 + 8 + 25 = \underline{\underline{48}}$$

⑧ How many integers between 1 and 10000 are divisible by 2, 3, 5, 7?

sol.

Peano Postulates

The Peano Postulates are a set of five axioms introduced by Italian mathematician G. Peano in 1889 to formally define and construct the natural numbers and their arithmetic properties.

1. Zero is a natural number, $0 \in \mathbb{N}$
2. Every natural number has a successor
for every $n \in \mathbb{N}$, there exists $S(n) \in \mathbb{N}$
s.t. $S(0) = 1, S(1) = 2, \dots$
3. Zero is not the successor of any natural number
for all $n \in \mathbb{N}$, $S(n) \neq 0$
4. Different numbers have different successors
If $S(m) = S(n)$ then $m = n$.
The successor function is one-to-one
5. Induction axiom

If a set $K \subseteq \mathbb{N}$ contains 0, and whenever $n \in K$ then $S(n) \in K$, then $K = \mathbb{N}$.

This is the whole basis of mathematical induction.

Mathematical Induction

Principle of mathematical induction

Let $P(n)$ be a proposition corresponding to the positive integers 'n' satisfying the following conditions

(i) $P(n_0)$ is true for some integer n_0 (Basis step)

(ii) If $P(k)$ is true for an arbitrary integer $k \geq n_0$, then

$P(k+1)$ is also true. Then $P(n)$ is true $\forall n \geq n_0$.

(Inductive step).

Problem 1

1. Prove by induction $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$, $n \geq 1$.

Sol:

Let $P(n) : 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$, $n \geq 1$

To prove $P(1)$ is true

$$1 = \frac{1(1+1)}{2} = 1$$

$\therefore P(1)$ is true

Assume that $P(k)$ is true for any ~~integer~~ ^{the integer} k .

i.e., $1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$

To prove $P(k+1)$ is true.

$$1 + 2 + 3 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$$

$$= (k+1) \left[\frac{k}{2} + 1 \right]$$

$$= (k+1) \left(\frac{k+2}{2} \right)$$

$$= \frac{(k+1)(k+2)}{2}$$

$\therefore P(k+1)$ is true.

Hence by the principle of mathematical induction $P(n)$ is true for all the integer n .

2. Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$, $n \geq 1$.

3. Prove the formula for the sum of first n cubes using mathematical induction.

4. Prove by mathematical induction that ~~2~~ $2^n > n$ for all $n \in \mathbb{N}$.

Sol:

Let $P(n): n < 2^n$

To prove $P(1)$ is true

$$1 < 2^1$$

$$\Rightarrow 1 < 2$$

$\therefore P(1)$ is true.

Assume that $P(k)$ is true

$$\text{i.e., } k < 2^k$$

To prove $P(k+1)$ is true

$$k < 2^k$$

$$\Rightarrow k+1 < 2^k + 1$$

$$\Rightarrow k+1 < 2^k + 2^k \quad \because [1 < 2^k]$$

$$\Rightarrow k+1 < 2^k (1+1)$$

$$\Rightarrow k+1 < 2^k \cdot 2$$

$$\Rightarrow k+1 < 2^{k+1}$$

$\therefore P(k+1)$ is true.

Hence $P(n)$ is true $\forall n \in \mathbb{N}$ by mathematical induction.

Ex. S.T. $a^n - b^n$ is divisible by $(a-b) \forall n \in \mathbb{N}$

Sol:

Let $P(n): a^n - b^n$ is divisible by $(a-b)$

To prove $P(1)$ is true

$$a^1 - b^1 = a - b$$

which is divisible by $(a-b)$

$\therefore P(1)$ is true

Assume that $P(k)$ is true

$$\text{i.e., } a^k - b^k = m(a-b)$$

$$\Rightarrow a^k = b^k + m(a-b) \longrightarrow \text{①}$$

(13)

To prove $P(k+1)$ is true.

$$\begin{aligned}\text{Now, } a^{k+1} - b^{k+1} &= a^k \cdot a - b^k \cdot b \\ &= a [b^k + m(a-b)] - b^k \cdot b \\ &= ab^k + am(a-b) - b^k \cdot b \\ &= b^k(a-b) + am(a-b) \\ &= (a-b)(b^k + am)\end{aligned}$$

which is divisible by $(a-b)$.

$\therefore P(k+1)$ is true.

Hence by the principle of mathematical induction $P(n)$ is true $\forall n \in \mathbb{N}$.

6. Use mathematical induction to prove that $n^3 - n$ is divisible by 3 $\forall n \in \mathbb{Z}^+$.

Sol:

Let $P(n): n^3 - n$ is divisible by 3.

To prove $P(1)$ is true

$$1^3 - 1 = 0 \text{ is divisible by 3}$$

$\therefore P(1)$ is true.

Assume that $P(k)$ is true

i.e. $k^3 - k$ is divisible by 3.

To prove $P(k+1)$ is true

$$\begin{aligned}(k+1)^3 - (k+1) &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= k^3 + 3k^2 + 3k - k \\ &= (k^3 - k) + 3(k^2 + k)\end{aligned}$$

which is divisible by 3.

$\therefore P(k+1)$ is true.

Hence $P(n)$ is true $\forall n \in \mathbb{Z}^+$.

Strong Induction

Principle of Mathematical Strong Induction

Let $P(n)$ be a proposition corresponding to the +ve integers 'n' satisfying the following conditions

(i) $P(n_0)$ is true for some integer n_0 .

(ii) If $P(k)$ is true for all integers $n_0 \leq k \leq n-1$, then $P(n)$ is also true. Then $P(n)$ is true $\forall n \geq n_0$.

7. Prove by mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each +ve integer.

Sol:

Let $P(n)$: $6^{n+2} + 7^{2n+1}$ is divisible by 43.

To prove $P(1)$ is true.

$$\begin{aligned} 6^{1+2} + 7^{2(1)+1} &= 6^3 + 7^3 \\ &= 216 + 343 \\ &= 559 \\ &= (43)(13) \end{aligned}$$

Which is divisible by 43.

$\therefore P(1)$ is true.

Assume that $P(k)$ is true

$$\text{i.e. } 6^{k+2} + 7^{2k+1} = 43m$$

To prove $P(k+1)$ is true.

$$\begin{aligned} 6^{(k+1)+2} + 7^{2(k+1)+1} &= 6^{k+3} + 7^{2k+3} \\ &= 6 \cdot 6^{k+2} + 7^2 \cdot 7^{2k+1} \\ &= 6 \cdot 6^{k+2} + (49) 7^{2k+1} \\ &= 6 \cdot 6^{k+2} + (43+6) 7^{2k+1} \end{aligned}$$

$$\begin{aligned} &= b \cdot b^{n+2} + b \cdot 7^{2k+1} + 43 \cdot 7^{2k+1} \\ &= b (b^{n+2} + 7^{2k+1}) + 43 \cdot 7^{2k+1} \\ &= b (42m) + 43 \cdot 7^{2k+1} \\ &= 43 [bm + 7^{2k+1}] \end{aligned}$$

which is divisible by 43

$\therefore P(k+1)$ is true

Hence $P(n)$ is true.

$b \cdot b^{n+2} + 7^{2n+1}$ is divisible by 43.

8. Any positive integer $n \geq 2$ is either a prime or a product of primes. To prove this by the principle of strong mathematical induction.

Sol:

Let $P(n)$: $n \geq 2$ is either a prime or a product of primes.

To prove $P(2)$ is true

$2 \geq 2$ is a prime

$\therefore P(2)$ is true

Assume that the statement is true for $2 \leq n \leq k$.

To prove $P(k+1)$ is true

For the integer $k+1$,

If $k+1$ is a prime, then the statement is true.

If $k+1$ is not a prime, then $k+1$ can be written as PQ , for some $2 \leq P, Q \leq k$.

According to the induction hypothesis, P and Q are either a prime or a product of primes. Consequently, PQ is a product of prime.